

# Seungwan Seo

Ph.D., in Industrial Engineering

## Contact Information

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### Robot Learning Research Scientist

CarbonSix  
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## Experience

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### Robot Learning Research Scientist

April 2026 -- Present

CarbonSix

- Develop VLA/foundation models for SigmaKit, from training to deployment
- Build, curate, and augment robot datasets
- Improve sim-to-real transfer for robust real-world robot performance

### Lead Vision Algorithm Engineer

May 2024 -- April 2026

GENGENAI

- Led the research and development of large-scale synthetic image generation pipelines for training and evaluating vision models
- Designed data generation frameworks for non-optical sensors (e.g., SAR and infrared), addressing extreme data scarcity scenarios
- Managed the end-to-end technical execution of client-driven synthetic data projects, from problem formulation to pipeline development

### Senior Researcher

February 2023 -- May 2024

AI Vision Cell, Hana Institute of Technology

- Conducted applied research on document understanding and OCR-based classification under real-world noisy conditions
- Developed generative models for virtual human synthesis, including facial expression and lip-motion generation from limited data
- Delivered a real-time virtual human prototype and demonstrated its feasibility at Korea Fintech Week 2023

## Education

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### Korea University

Ph.D. in Industrial Management

Seoul

2017 - 2023

### Seoul Tech

Bachelor of Science in Information Technology and Management

Seoul

2011 - 2017

## Selected Projects (Industry)

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@ CARBONSIX

2026.04 – CURRENT

### Develop CarbonSix's proprietary robot foundation model

2025.08 -- Present

CarbonSix, Research

- Train the model from scratch rather than adapting existing open-source WAMs or VLAs

## Enhanced ML capabilities in CarbonSix's robot learning system

2025.04 -- Present

CarbonSix, Research

- Introduced Diffusion and Flow Matching for action generation
- Implemented real-time action chunking for smoother robot motion without performance degradation, reducing jerk by over 70%
- Introduced latent-space feature prediction to improve model precision, achieving over a 7 percentage-point accuracy gain on real-world tasks

## @ GENGENAI

2024.05 – 2026.04

### High-altitude Narrow-FOV Synthetic Image Generation

2025.05 -- 2026.04

Korea Aerospace Industries (Client)

- Generated synthetic imagery simulating observations from altitudes up to 9,000 m (30,000 ft) under extremely narrow field-of-view constraints
- Generated full scenes containing TEL and SAM assets using conditional T2I models, with precise control over viewing geometry and object scale
- Enabled configurable camera tilt (pitch) angles ranging from 15° to 90° to support diverse observation scenarios
- Modeled diverse environmental conditions (e.g., snow, rain, fog) to support data generation across varied weather scenarios

### Drone-view Synthetic Data Generation via Conditional T2I Models

2025.01 -- 2025.05

Korea Aerospace Industries (Client)

- Generated drone-view image datasets using conditional text-to-image models, leveraging open datasets such as VisDrone
- Synthesized drone-view scenes with precise spatial control over the placement and type of TEL assets
- Achieved a 7 p.p. improvement in object detection performance (mAP@50) on real drone-captured test images when training exclusively on synthetic data, compared to models trained on open datasets
- Reduced generation latency by replacing inpainting-based pipelines with condition-controlled T2I generation
- Trained detection models capable of identifying TEL decoys with up to 90% detection accuracy (mAP@50)

### Satellite-view Synthetic Image Generation for TEL Detection

2024.10 -- 2024.12

Korea Aerospace Industries (Client)

- Generated nadir-view (top-down) satellite imagery with controllable placement of specific TEL variants
- Designed scene-consistent synthetic data pipelines to replicate realistic satellite observation conditions
- Produced fully labeled datasets optimized for object detection under limited availability of real satellite imagery

### Core Research on Synthetic Data Generation for Autonomous Driving

2024.05 - 2024.10

GENGENAI, Foundational Research

- Developed scene-aware inpainting techniques to insert vehicles, cyclists, and pedestrians at controllable locations within autonomous driving scenes
- Mitigated severe class imbalance in object detection datasets by selectively synthesizing under-represented object categories
- Validated the effectiveness of the synthetic data through a 16% improvement in object detection performance (mAP@50) compared to training on real-world datasets alone

## @HANA INSTITUTE OF TECHNOLOGY

2023.02 – 2024.05

### Virtual Human Generation from Limited Visual Data

2023.04 - 2024.05

Hana Institute of Technology, Foundational Research

- Developed generative pipelines for virtual human synthesis using diffusion and StyleGAN-based models
- Generated identity-consistent virtual humans from limited visual inputs, including single-image scenarios
- Implemented facial attribute manipulation and lip motion generation conditioned on user-provided scripts
- Built an end-to-end system for producing talking-head videos, enabling scalable content generation for VTM applications

## OCR-based Document Classification for Financial Applications

2023.02 - 2023.05

Hana Bank

- Designed an OCR-based document understanding system for foreign exchange and financial documents
- Addressed classification challenges arising from handwritten text, noisy scans, and heterogeneous document layouts
- Classified document types into fine-grained categories and extracted key phrases to support downstream financial processing

## Selected Projects (Doctoral Research)

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### Synthetic Data Generation for Unseen Ocean Environments

2019.09 - 2020.05

Daewoo Shipbuilding & Marine Engineering

- Generated synthetic ocean imagery using GANs to improve classification performance on unseen data

### Ocean Wave Property Prediction from Imagery

2020.12 - 2021.10

Daewoo Shipbuilding & Marine Engineering

- Developed deep learning-based models to predict wave height, direction, and period from ocean wave images

### DeepFake Video Detection

2019.03 - 2021.02

National Research Foundation of Korea

- Developed DeepFake detection models using classification and anomaly detection techniques

### Sentence Similarity Estimation via Image Association

2018.06 - 2019.02

NCSOFT

- Designed multimodal models to estimate sentence similarity through image-based associative representations

### System Anomaly Detection Using Deep Learning

2018.03 - 2019.03

Agency for Defense Development

- Detected abnormal system states using RNN-based autoencoder models

## Publications

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### Exploring the Differences in Adversarial Robustness Between ViT- and CNN-Based Models Using Novel Metrics

2023.10

Seungwan Seo, Jaehyuk Heo, Pilsung Kang\*

Computer Vision and Image Understanding, IF: 4.886, Q1 Journal

[Original paper link](#)

### Evaluating and improving adversarial robustness of CNNs on image classification

2023.02

Seungwan Seo

Korea University

[Original paper link](#)

### Cost-free Adversarial Defense: Distance-based optimization for model robustness without adversarial training

2023.01

Seungwan Seo, Yunseung Lee, Pilsung Kang\*

Computer Vision and Image Understanding, IF: 4.886, Q1 Journal

[Original paper link](#)

### Intrusion Detection based on Sequential Information preserving Log Embedding Methods and Anomaly Detection Algorithms

2021.04

Czangyeob Kim, Myeongjun Jang, Seungwan Seo, Kyeongchan Park, Pilsung Kang\*

IEEE Access, IF: 4.34, Q1 Journal

[Original paper link](#)

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| <b>Unusual customer response identification and visualization based on text mining and anomaly detection</b><br>Seungwan Seo, Deokseong Seo, Myeongjun Jang, Jaeyun Jeong, Pilsung Kang*<br>Expert Systems with Applications, Volume 144, April 2020, 113111, IF: 9.60, Q1 Journal<br><a href="#">Original paper link</a>              | 2020.04 |
| <b>Comparative Study of Deep Learning-Based Sentiment Classification</b><br>Seungwan Seo, Czangyeob Kim, Haedong Kim, Kyoung Hyun Mo, Pilsung Kang*<br>IEEE Access, Volume 8, January 2020, Pages 6861 - 6875, IF: 4.34, Q1 Journal<br><a href="#">Original paper link</a>                                                             | 2020.01 |
| <b>Recurrent neural network-based semantic variational autoencoder for Sequence-to-sequence learning</b><br>Myeongjun Jang, Seungwan Seo, Pilsung Kang*<br>Information Sciences, Volume 490, July 2019, Pages 59-73, IF: 6.795, Q1 Journal<br><a href="#">Original paper link</a>                                                      | 2019.07 |
| <b>Sentiment classification with word localization based on weakly supervised learning with a convolutional neural network</b><br>Gichang Lee, Jaeyun Jeong, Seungwan Seo, CzangYeob Kim, Pilsung Kang*<br>Knowledge-Based Systems Volume 152, 15 July 2018, Pages 70-82, IF: 8.038, Q1 Journal<br><a href="#">Original paper link</a> | 2018.07 |

## Conference

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| <b>An Active Reference Reset Method Adapting Distribution Shift for Robust System Anomaly Detection</b><br>Seungwan Seo, Heejeong Choi, Pilsung Kang*<br>2022 13th International Conference on Information and Communication Technology Convergence (ICTC)<br><a href="#">Original paper link</a> | 2022.10 |
| <b>Cost-free Adversarial Defense: Distance-based optimization for model robustness without adversarial training</b><br>Korean Institute of Industrial Engineers (Spring), Jeju Convention Center, Jeju<br>Seungwan Seo, Yunseng Lee, Pilsung Kang*                                                | 2021.06 |
| <b>Intrusion detection based on sequential information preserving log embedding methods and anomaly detection algorithms</b><br>INFORMS Annual Meeting, Seattle, WA<br>Seungwan Seo, Pilsung Kang*                                                                                                | 2019.11 |
| <b>Measuring Sentence Similarity based on Image Association and Siamese Network</b><br>Korean Institute of Industrial Engineers (Fall), Seoul National University, Seoul<br>Seungwan Seo, Minsung Jeong, Heejeong Choi, Pilsung Kang*                                                             | 2019.11 |
| <b>Identifying and visualizing uncommon customer response on machine learning</b><br>INFORMS Annual Meeting, Phoenix, Arizona<br>Seungwan Seo, Pilsung Kang*                                                                                                                                      | 2018.11 |
| <b>Comparative Study of Deep Learning-Based Sentiment Classification</b><br>Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju<br>Seungwan Seo, Czangyeob Kim, Haedong Kim, Kyoung Hyun Mo, Pilsung Kang*                                                                 | 2018.04 |

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| <b>Unusual customer response identification and visualization based on text mining and anomaly detection</b><br>Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju<br>Seungwan Seo, Deokseong Seo, Myeongjun Jang, Jaeyun Jeong, Pilsung Kang*     | 2018.04 |
| <b>Distance Decomposition for Variable Importance of Distance-based Novelty Detection</b><br>Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju<br>Seungwan Seo, Myeongjun Jang, Czangyeob Kim, Pilsung Kang*                                      | 2018.04 |
| <b>Recurrent Neural Network-based Semantic Variational Autoencoder for Sequence to Sequence Learning</b><br>Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju<br>Seungwan Seo, Myeongjun Jang, Pilsung Kang*                                      | 2018.04 |
| <b>Sentiment classification with word localization based on weakly supervised learning with a convolutional neural network</b><br>Korean Institute of Industrial Engineers (Fall), KAIST, Daejeon<br>Seungwan Seo, Gichang Lee, Jaeyun Jeong, Czangyeob Kim, Pilsung Kang* | 2017.11 |

## Teaching Experiences

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| <b>Deep Learning Course</b><br>LG Innotek<br>Full-time Lecture                                          | 2022.05       |
| <b>Python Syntax Course</b><br>LG Innotek<br>Full-time Lecture                                          | 2022.05       |
| <b>Python for Deep Learning Course</b><br>Fastcampus<br>Full-time Lecture                               | Online course |
| <b>Python Syntax Course</b><br>LG Innotek<br>Full-time Lecture                                          | 2021.07       |
| <b>Python for Machine Learning and Deep Learning Course</b><br>SK Hynix<br>Full-time Lecture            | 2020.11       |
| <b>Python for Machine Learning and Deep Learning Course</b><br>SK Hynix<br>Full-time Lecture            | 2020.03       |
| <b>Machine Learning with Python</b><br>Korea Credit Bureau<br>Full-time Lecture                         | 2019.09       |
| <b>Machine Learning with Python</b><br>Korea Credit Bureau<br>Full-time Lecture                         | 2019.08       |
| <b>Python for Machine Learning and Deep Learning Course</b><br>Samsung Electronics<br>Full-time Lecture | 2019.07       |
| <b>Python for Machine Learning and Deep Learning Course</b><br>Samsung Electronics<br>Full-time Lecture | 2019.07       |

**Python for Deep Learning for Text Domain**

SeoulTech  
*Seminar*

2019.02

**Python for Machine Learning and Deep Learning Course**

Samsung Electronics  
*Full-time Lecture*

2019.02

**Python Syntax Course**

Samsung Electronics  
*Full-time Lecture*

2019.01